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FIGURE 1. Multiple hematomas located at the section of spleen.

Spontaneous rupture of the spleen, although rare, can also occur. Accompanying the pathological process, splenic rupture can happen with or without minor trauma. In forensic practice, forensic pathologists must make careful search and thorough investigation to make a differential diagnosis between traumatic and spontaneous splenic rupture.

The clinical progression of NPD-B is slow, with no or only mild neurological manifestations. Patients can live with the disease for many years and develop solitary splenomegaly in their later years.⁸ Splenomegaly is the most common clinical manifestation of NPD, accounting for 99% of the clinical manifestations. In this case, no other clinical symptoms were seen except the slight splenomegaly; only the pathological examination revealed the presence of Niemann-Pick cells.

It is well known that many diseases states can make a spleen enlarged and more susceptible to injury. A previous report also indicated spontaneous rupture of the spleen can occur in NPD.⁹ A 27-year-old woman was admitted to the hospital for spontaneous splenic rupture and underwent splenectomy. The medical history showed chronic anemia and splenomegaly. Pathological examinations of spleen and bone marrow biopsy showed foam cells contributed to the diagnosis of NPD. She recovered well after the surgery. In 2008, Chen et al¹⁰ reported another spontaneous rupture caused by NPD-B, a 53-year-old man who was finally diagnosed after splenectomy. They also reviewed other 4 similar cases and concluded the man in their case was the oldest. All of the reported cases occurred in adults, and all of them were diagnosed postsurgery. In 2016, Yao et al¹¹ reported the case of a 66-year-old woman finally diagnosed as having NPD-B after splenectomy because of persistent pain in the left upper abdomen. Although splenomegaly was common in this type of NPD, spontaneous splenic rupture rarely happens.

The mechanisms of spontaneous splenic rupture with NPD may be due to the enlarged spleen and accumulation of lipids; whether the altered structure makes sense still needs further research. In present case, slight splenomegaly and abnormal hypertrophic foam cells (Niemann-Pick cells) were found around the splenic corpuscle and in the spleen sinus, with a focus on hemosiderosis, indicating the spleen may have been more susceptible to injury. Considering the trauma history, it is likely that the trauma and disease together resulted in the rupture of spleen. Because of the presence of underlying NPD-B in the victim, the police decided not to prosecute the suspect for assault in this case (Table 1).

CONCLUSIONS

Splenic rupture is not uncommon in clinical and forensic practice, and although in the majority of cases the cause is trauma,

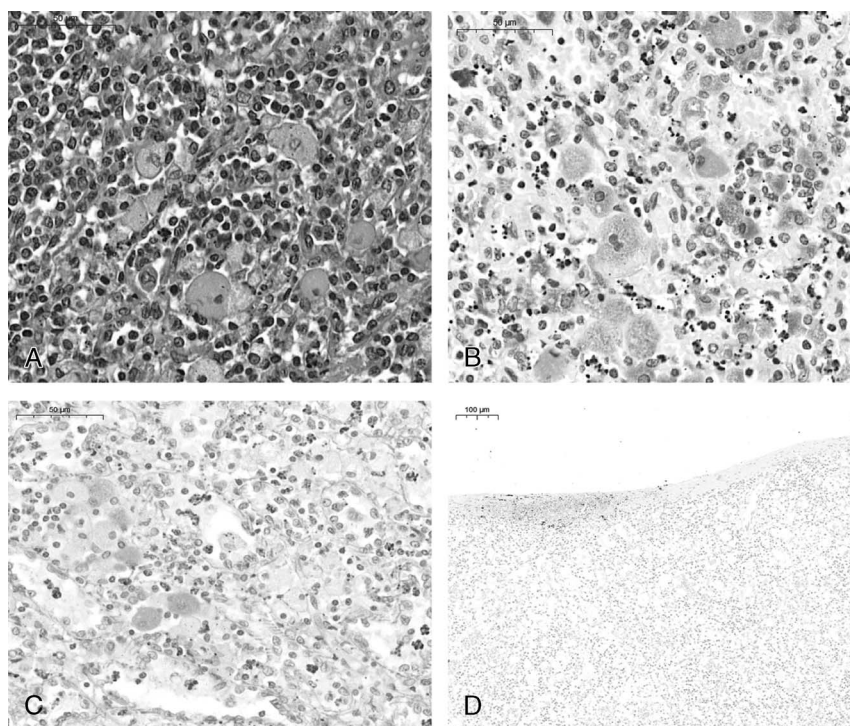


FIGURE 2. Niemann-Pick cells were found around the splenic corpuscle and in the spleen sinus with a small piece of hemosiderosis under the splenic capsule: (A) hematoxylin-eosin stain, original magnification $\times 400$; (B) CD68, original magnification $\times 400$; (C) PAS, original magnification $\times 400$; (D) Prussian blue, original magnification $\times 100$.

TABLE 1. Classifications of NP Disease

	NPD-A	NPD-B	NPD-C
Cause	ASM deficiency	ASM deficiency	Cholesterol transport disorder
Gene	SMPD1	SMPD1	NPC1/2
Clinical characteristics	Cherry red macula lutea and corneal opacity	No or only mild neurological manifestation	Vertical supranuclear gaze palsy
Outcome	Few patients survive >4 y old	Can survive with solitary splenomegaly	Cognitive dysfunction

spontaneous rupture of the spleen can also occur. In forensic practice, forensic pathologists must perform a thorough investigation as to the cause of the rupture, including microscopy to differentiate between traumatic and spontaneous rupture, as the abnormal of microscopic examination may indicate the possibility of an underlying disease process, and additional studies, including genetic testing, may be indicated.

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